Full stack Engineer – Technical Assessment

Introduction

Congratulations – you've made it to the next step in the process for the role as Full stack Engineer at Svea Solar! We've set up a technical assessment for you to ensure that we're a good match. What we're looking for is your ability to reason around this task and how you structure your solution.

We don't expect you to complete all parts to 100% – what matters most is the conversation we'll have around your code, design choices, and trade-offs.

Overview

This assessment evaluates your proficiency with React, TypeScript, and full stack development practices relevant to the role of the Sun Sites team at Svea Solar.

You'll be building a small "Solar Quote Platform" that includes a calculator, a lead capture form, and a simple backend service to store and retrieve submitted quotes.

Please track how much time you spend on the assignment (we don't expect you to work on it for more than a few hours)

Requirements

Tech Stack

We'd like to see React and Typescript for the frontend. You are free to pick and choose whatever approach you like using them. Backend, we will leave it to you. Regardless, we expect you to be able to explain how you've picked your stack and why you think it fits the project.

Artificial Intelligence

We love AI. And we hope you do too. The only requirement if you are using AI to complete your task is that you must come prepared to tell us all about your process. 😉

Case: Solar Quote Platform

We want you to build a solar quotation platform.

Application Features

1. Quote form page

A page where customers can calculate their potential savings and send in their contact information to get a personalized quote. We envision two sections:

Section 1 – Savings calculations

A way for the customer to see the potential of our offering. We think it should:

- Let customers input average monthly energy bill
- Visualize the monthly savings

*- You can hardcode or mock the calculation logic (for example, savings = bill * 0.3).*

Section 2 - Lead Capture Form

A contact form that lets the customer send a request for a personalized quote. We think it should:

- Capture essential information (name, email, address, etc.)
- Send data through to the backend API

2. Backend API

A small backend API that handles the lead submissions and stores them in some way. We think the API simply needs:

- To receive quote requests
- To return all the quote requests

3. Quote Dashboard

The last part of the application is a quote dashboard. This page is meant as a back-office view of the submitted quote requests. We think it should:

- Fetch and display all requested quotes in some sort of list

Evaluation Criteria

- Code quality and organization
- Architectural choices
- Hard skills (TypeScript, React, Styling, etc…)
- Reasoning around implementation choices

Submission Guidelines

1. Create a new repository on GitHub
2. Implement the application according to the requirements
3. Include a README with:
    a. Setup instructions
    b. Brief explanation of implementation choices
4. Submit the GitHub repository URL

Let us know if you have any questions. Good luck!